

Hypothesis Testing with E-Values

Chapter 14: Further Contrasts between E-Values and P-Values

Based on Ramdas & Wang

Outline

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Recap: e-variables and p-variables I

Intuition. Before contrasting the two objects further, let us recall exactly what they are. Both are data-dependent numbers meant to quantify evidence against a null hypothesis, but they are calibrated in opposite directions: an e-value is *large* when there is evidence against H_0 , a p-value is *small*.

Definitions (from earlier chapters)

Let $\mathcal{P}$ be a set of probability measures (the null hypothesis) on a sample space. A random variable $E \geq 0$ is an **e-variable** for $\mathcal{P}$ if

$$\mathbb{E}^{\mathbb{P}}[E] \leq 1 \quad \text{for every } \mathbb{P} \in \mathcal{P}.$$

A random variable $P \in [0, 1]$ is a **p-variable** for $\mathcal{P}$ if

$$\mathbb{P}(P \leq \alpha) \leq \alpha \quad \text{for every } \mathbb{P} \in \mathcal{P}, \alpha \in (0, 1).$$

An e-variable is **exact** if $\mathbb{E}^{\mathbb{P}}[E] = 1$ for every $\mathbb{P} \in \mathcal{P}$; a p-variable is **exact** if P is uniformly distributed on $[0, 1]$ under every $\mathbb{P} \in \mathcal{P}$.

Recap: e-variables and p-variables II

What this chapter adds. Chapter 14 sharpens the relationship between these two objects along three fronts:

- **14.1** A precise *duality*: the best possible e-value and the best possible p-value, built from the same monotone test statistic, are exact reciprocals of one another.
- **14.2** A *structural* fact: every e-variable is (dominated by) a conditional expectation of a likelihood ratio, and every p-variable is (dominated by) a conditional probability. This is literally why we write “e” for e-value.
- **14.3** An *existence* result: for finite hypotheses, exact e-values/p-values that are powered against an alternative exist iff a geometric separation condition on $\text{Span}(\mathcal{P})$ holds.

14.1 Setup: monotone test statistics I

Intuition. In many tests we do not use an arbitrary e- or p-variable: we compute a real-valued test statistic X (e.g. a sample sum) where larger X means more evidence against H_0 . It is then natural to require the e-variable to be an *increasing* function of X , and the p-variable to be a *decreasing* function of X – both should move the same direction as “evidence”.

Left-continuous survival function

For $x \in \mathbb{R}$, define

$$S(x; \mathbb{P}) = \mathbb{P}(X \geq x).$$

Here X is the test statistic, $\mathbb{P} \in \mathcal{P}$ is a candidate null distribution, and $S(\cdot; \mathbb{P})$ is the (left-continuous) survival function of X under $\mathbb{P}$: the chance of seeing evidence at least as extreme as x .

14.1 Setup: monotone test statistics II

Why $S(X; \mathbb{P})$ matters. It behaves like an “idealized p-value”: it is exactly the tail probability of the observed statistic. Proposition 14.1 shows it sandwiches *every* monotone e- and p-variable.

Proposition 14.1

Suppose E is an e-variable for $\mathbb{P}$ that is an increasing function of X , and P is a p-variable for $\mathbb{P}$ that is a decreasing function of X . Then, $\mathbb{P}$ -almost surely,

$$P \geq S(X; \mathbb{P}) \quad \text{and} \quad E \leq \frac{1}{S(X; \mathbb{P})}.$$

Symbols: P = any monotone p-variable; E = any monotone e-variable; $S(X; \mathbb{P})$ = the survival function evaluated at the realized statistic X .

14.1 Setup: monotone test statistics III

Step-by-step derivation (p-variable half).

- ① Let T be any decreasing function and X' an i.i.d. copy of X . Because T is decreasing, $T(X') \leq T(X) \iff X' \geq X$ is not quite right in general, but the relevant *inclusion* holds:

$$\{T(X') \leq T(X)\} \supseteq \{X' \geq X\}.$$

- ② Taking conditional probabilities given X (using that $X' \sim X$ under $\mathbb{P}$),

$$\mathbb{P}(T(X') \leq T(X) \mid X) \geq \mathbb{P}(X' \geq X \mid X) = S(X; \mathbb{P}) \quad \mathbb{P}\text{-a.s.}$$

- ③ A p-variable built as $P = T(X)$ satisfies, by definition, that its cdf F_P under $\mathbb{P}$ is $\leq$ the identity on $[0, 1]$, hence $P \geq F_P(P) \geq \mathbb{P}(T(X') \leq T(X) \mid X) \geq S(X; \mathbb{P})$.

E-variable half. Follows immediately: $1/E$ is itself a p-variable (a standard calibration fact), decreasing in X since E is increasing; apply the p-variable bound to $1/E$ to get $1/E \geq S(X; \mathbb{P})$, i.e. $E \leq 1/S(X; \mathbb{P})$.

Theorem 14.2: The duality I

Intuition. Proposition 14.1 gives a one-sided bound: $S(X; \mathbb{P})$ is an upper bound on the best e-value's reciprocal. The next theorem shows this bound is *tight* – it is not merely an inequality but an exact identity between suprema and infima. This is the “duality” advertised in the section title.

Theorem 14.2 (Duality between e-values and p-values)

Suppose the measures in $\mathcal{P}$ are absolutely continuous w.r.t. a reference measure $\mathbb{L}$. Let $\mathcal{E}^X$ be the set of all e-variables for $\mathcal{P}$ that are increasing functions of X , and $\mathcal{U}^X$ the set of all p-variables for $\mathcal{P}$ that are decreasing functions of X . Then, $\mathbb{L}$ -almost surely,

$$\sup_{E \in \mathcal{E}^X} E = \inf_{\mathbb{P} \in \mathcal{P}} \frac{1}{S(X; \mathbb{P})} = \frac{1}{\inf_{P \in \mathcal{U}^X} P}.$$

In particular, if $\mathcal{P} = \{\mathbb{P}\}$ is a single distribution, then $\sup_{E \in \mathcal{E}^X} E = 1/S(X; \mathbb{P})$, $\mathbb{P}$ -almost surely.

Symbols: $\mathcal{E}^X, \mathcal{U}^X$ = all monotone e-/p-variables built from X ; the middle term is the tightest possible survival bound over the whole null $\mathcal{P}$.

Theorem 14.2: The duality II

Step-by-step derivation.

- ① Proposition 14.1 already gives one inequality chain:

$$\sup_{E \in \mathcal{E}^X} E \leq \sup_{P \in \mathcal{U}^X} \frac{1}{P} \leq \inf_{\mathbb{P} \in \mathcal{P}} \frac{1}{S(X; \mathbb{P})} \quad \mathbb{L}\text{-a.s.} \quad (14.1)$$

- ② *Achievability.* For each threshold $s \in \mathbb{R}$, construct

$$E_s = \frac{1}{\sup_{\mathbb{P} \in \mathcal{P}} S(s; \mathbb{P})} \mathbf{1}\{X \geq s\} \quad (E_s := 1 \text{ if } 0/0).$$

This E_s is increasing in X (an indicator of an upper set).

- ③ *Check it is a valid e-variable.* For any $\mathbb{P} \in \mathcal{P}$,

$$\mathbb{E}^{\mathbb{P}}[E_s] = \frac{S(s; \mathbb{P})}{\sup_{\mathbb{P}' \in \mathcal{P}} S(s; \mathbb{P}')} \leq 1.$$

Theorem 14.2: The duality III

- ④ *Optimize over the threshold.* Plugging in $s = X$ (i.e. taking the supremum over $s \in \mathbb{R}$ of E_s) exactly achieves

$$\sup_{s \in \mathbb{R}} E_s = \frac{1}{\sup_{\mathbb{P} \in \mathcal{P}} S(X; \mathbb{P})}.$$

- ⑤ This matches the right-hand side of (14.1), so the chain of inequalities collapses to equalities.

Reading. The theorem says the reciprocal of the “worst-case” survival probability is not just an upper bound – it is literally *achieved* by an explicit e-variable, and it equals the reciprocal of the best possible p-variable. E-values and p-values, restricted to monotone transforms of one statistic, are two faces of the same extremal object.

Numerical illustration: $e = 4 \leftrightarrow p = 0.25$

Concrete setup. Take $\mathcal{P} = \{\mathbb{P}\}$ a single null, and suppose X has survival function $S(x; \mathbb{P}) = e^{-x}$ (e.g. X is Exponential(1) under $\mathbb{P}$) – a simple monotone test statistic where larger x is more extreme.

Arithmetic

Suppose we observe $X = x_0$ where the tail probability works out to

$$p := S(x_0; \mathbb{P}) = 0.25.$$

By Theorem 14.2, the best (largest) e-value obtainable from this same statistic is exactly the reciprocal:

$$e := \sup_{E \in \mathcal{E}^X} E = \frac{1}{S(x_0; \mathbb{P})} = \frac{1}{0.25} = 4.$$

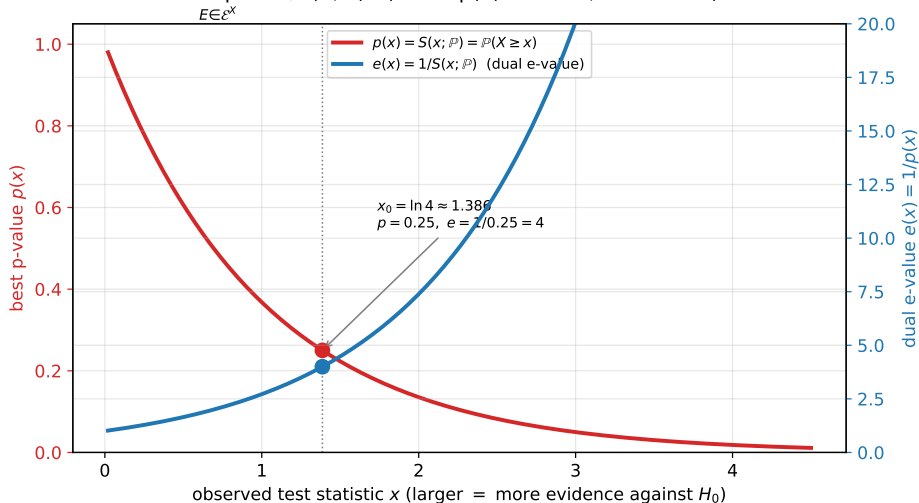
Concretely, since $S(x; \mathbb{P}) = e^{-x}$, solving $e^{-x_0} = 0.25$ gives $x_0 = -\ln(0.25) \approx 1.386$.

Reading the numbers. A p-value of 0.25 and an e-value of 4 are not two independently calibrated notions of “mild evidence” – under the duality map they are *the same piece of information*, viewed through the reciprocal $e = 1/p$. This is exactly the curve plotted in Figure 1 (next frame’s companion figure), where the red p-value curve $S(x)$ and blue e-value curve $1/S(x)$ cross through the marked point $(x_0, p=0.25) \leftrightarrow (x_0, e=4)$.

Figure: the duality curve

Theorem 14.2: the duality between e-values and p-values

$\sup_{E \in \mathcal{E}^X} E = 1/S(X; \mathbb{P})$ ($X \sim \text{Exp}(1)$ under $\mathbb{P}$, illustrative)



Python: the duality map numerically

```
import numpy as np

rng = np.random.default_rng(0)

def S(x):                                # survival fn of  $X \sim \text{Exp}(1)$  under  $P$ 
    return np.exp(-x)

# --- duality map  $e(x) = 1 / S(x;P)$ , for several observed  $x$  ---
for x in [0.5, 1.0, 1.386, 2.0, 3.0]:
    p = S(x)                             # best (smallest) p-value at this  $x$ 
    e = 1.0 / p                           # dual (largest) e-value, Theorem 14.2
    print(f"x={x:5.3f}  p={p:.4f}  e=1/p={e:8.3f}")

# --- concrete check:  $e = 4 \iff p = 0.25$  ---
x0 = -np.log(0.25)                       # solve  $\exp(-x_0) = 0.25$ 
print(f"x0 with p=0.25:", round(x0, 4), " -> e =", 1 / 0.25)

# --- Monte Carlo check of Corollary 14.3 (Ville/Markov bound) ---
N = 200_000
X = rng.exponential(scale=1.0, size=N)
alpha = 0.1
E = 1.0 / S(X)                           # the extremal e-variable  $\sup_s E_s$ 
empirical = np.mean(E >= 1 / alpha)
print(f"P(E >= 1/alpha) empirical:", round(empirical, 4),
      f"  <= alpha =", alpha)
```

Output confirms $e(x_0=1.386) \approx 4.00$ and the empirical tail probability sits at (or just below) $\alpha = 0.1$, consistent

Corollary 14.3: a strengthened Ville/Markov bound I

Intuition. Ordinary Markov's inequality for a single e-variable E says $\mathbb{P}(E \geq 1/\alpha) \leq \alpha$. Because the duality theorem produces an explicit *extremal* e-variable dominating an entire family $\mathcal{E}$ at once, we get a uniform version: the supremum over a whole comonotonic family of e-variables still only breaches $1/\alpha$ with probability at most α .

Corollary 14.3

For any set $\mathcal{E}$ of e-variables for $\mathcal{P}$ that are increasing functions of X ,

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P} \left(\sup_{E \in \mathcal{E}} E \geq \frac{1}{\alpha} \right) \leq \alpha \quad \text{for all } \alpha \in (0, 1).$$

Symbols: $\mathcal{E}$ = any collection of monotone e-variables (not just one); the bound holds for the *supremum* of the whole collection simultaneously, uniformly over $\mathbb{P} \in \mathcal{P}$.

Corollary 14.3: a strengthened Ville/Markov bound II

Why this is stronger than plain Markov. Applying Markov's inequality to each $E \in \mathcal{E}$ separately and union-bounding over $|\mathcal{E}|$ elements would cost a factor of $|\mathcal{E}|$ (or require a Bonferroni correction). Corollary 14.3 needs no such correction: because every $E \in \mathcal{E}^X$ is dominated $\mathbb{P}$ -a.s. by the single random variable $1/S(X; \mathbb{P})$ (Theorem 14.2), the event $\{\sup_E E \geq 1/\alpha\}$ is itself controlled by the tail of $S(X; \mathbb{P})$, and no union bound is needed.

The key structural reason: comonotonicity. What actually matters for this “free lunch” is that all e-variables under consideration are *comonotonic* – increasing functions of the *same* random variable X (or all decreasing in X ; monotonicity can be flipped throughout by using the right-continuous survival function instead). Chapter 15.2 develops this comonotonicity idea more generally.

14.2 Motivation: what is an e-value, structurally?

Intuition. So far, e-variables have been defined by an *inequality*: $\mathbb{E}^{\mathbb{P}}[E] \leq 1$. This section shows something stronger and more structural: every e-variable is (dominated by) an actual **conditional expectation** of a likelihood ratio $d\mathbb{Q}/d\mathbb{P}$ given the data. Likewise, every p-variable is (dominated by) an actual **conditional probability**. This is a deeper fact than the defining inequalities alone – it explains why the letter “e” stands for *expectation*.

Setup. Let X be a generic data sample (a measurable map into $\mathbb{R}^n$; the ambient space plays no role here). A p- or e-variable Y is *built on* X if it is a measurable function of X – i.e. Y only depends on the data through X .

Theorem 14.4: e-variables are conditional expectations I

Theorem 14.4: e-variables are conditional expectations II

Theorem 14.4 (Representation for e-variables)

For a random variable E , the following are equivalent:

- (i) E is an e-variable for $\mathcal{P}$ built on X ;
- (ii) for every $\mathbb{P} \in \mathcal{P}$, there exists a probability measure $\mathbb{Q} \ll \mathbb{P}$ such that

$$E \leq \mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \mid X \right] \quad \mathbb{P}\text{-almost surely.} \quad (14.2)$$

If all $\mathbb{P} \in \mathcal{P}$ are equivalent to some measure $\mathbb{L}$, every e-variable built on X is dominated by

$$\inf_{\mathbb{P} \in \mathcal{P}} \mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}_{\mathbb{P}}}{d\mathbb{P}} \mid X \right],$$

itself an e-variable for $\mathcal{P}$, for some class $(\mathbb{Q}_{\mathbb{P}})_{\mathbb{P} \in \mathcal{P}}$ absolutely continuous w.r.t. $\mathbb{L}$. *Symbols:* $d\mathbb{Q}/d\mathbb{P}$ is the Radon–Nikodym derivative (likelihood ratio) of an *alternative* $\mathbb{Q}$ against the null $\mathbb{P}$; conditioning on X averages this ratio over whatever additional randomness is not captured by X .

Theorem 14.4: e-variables are conditional expectations III

A familiar special case. When densities exist (Chapter 5), the conditional expectation in (14.2) often reduces to

$$\mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \mid X \right] = \frac{p(X)}{q(X)},$$

where p, q are the densities of X under $\mathbb{P}, \mathbb{Q}$: exactly the ordinary *likelihood ratio* test statistic.

Step-by-step derivation.

① $(ii) \Rightarrow (i)$. By the tower property of conditional expectation, for every $\mathbb{P} \in \mathcal{P}$,

$$\mathbb{E}^{\mathbb{P}}[E] \leq \mathbb{E}^{\mathbb{P}} \left[\mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \mid X \right] \right] = \mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \right] = 1.$$

So E satisfies the e-variable inequality for every $\mathbb{P}$.

② $(i) \Rightarrow (ii)$. Fix $\mathbb{P} \in \mathcal{P}$. If $\mathbb{E}^{\mathbb{P}}[E] > 0$, define $\mathbb{Q}$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \frac{E}{\mathbb{E}^{\mathbb{P}}[E]};$$

this is a valid density since it is ≥ 0 with $\mathbb{P}$ -expectation 1. If $\mathbb{E}^{\mathbb{P}}[E] = 0$, just set $\mathbb{Q} = \mathbb{P}$.

Theorem 14.4: e-variables are conditional expectations IV

- 8 Verify (14.2) holds with equality-or-more. For $\mathbb{P}$ with $\mathbb{E}^{\mathbb{P}}[E] > 0$,

$$\mathbb{E}^{\mathbb{P}}\left[\frac{d\mathbb{Q}}{d\mathbb{P}} \mid X\right] = \mathbb{E}^{\mathbb{P}}\left[\frac{E}{\mathbb{E}^{\mathbb{P}}[E]} \mid X\right] = \frac{E}{\mathbb{E}^{\mathbb{P}}[E]} \geq E \quad (\text{since } \mathbb{E}^{\mathbb{P}}[E] \leq 1).$$

For $\mathbb{P}$ with $\mathbb{E}^{\mathbb{P}}[E] = 0$, $E = 0$ $\mathbb{P}$ -a.s., and (14.2) holds trivially since the right side is $\geq 0 = E$.

Takeaway. Every e-value *is*, structurally, a (bounded version of a) conditional expectation of some likelihood ratio – not merely a number satisfying an expectation constraint. Theorem 14.4 generalizes the simple-null likelihood-ratio connection from Proposition 2.12 to arbitrary composite $\mathcal{P}$.

Theorem 14.5: p-variables are conditional probabilities I

Intuition. The mirror-image fact holds for p-values: a p-variable is (dominated by) the conditional probability that an independent replicate of the data looks “at least as extreme” as what was observed – precisely capturing the everyday informal description of a p-value.

Theorem 14.5: p-variables are conditional probabilities II

Theorem 14.5 (Representation for p-variables)

For a random variable P , the following are equivalent:

- (i) P is a p-variable for $\mathcal{P}$ built on X ;
- (ii) there exists a measurable $T : \mathbb{R}^n \rightarrow \mathbb{R}$ such that, for every $\mathbb{P} \in \mathcal{P}$,

$$P \geq \mathbb{P}(T(X') \leq T(X) \mid X) \quad \mathbb{P}\text{-a.s.}, \quad (14.3)$$

where X' is an i.i.d. copy of X under $\mathbb{P}$.

If all $\mathbb{P} \in \mathcal{P}$ are equivalent to some $\mathbb{L}$, every p-variable built on X is dominated by (i.e. $\leq$)

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(T(X') \leq T(X) \mid X),$$

itself a p-variable for $\mathcal{P}$. *Symbols:* T = a test statistic (possibly different from X itself); X' = a fresh independent draw under the *same* $\mathbb{P}$ used for conditioning.

Theorem 14.5: p-variables are conditional probabilities III

Step-by-step derivation.

- ① $(ii) \Rightarrow (i)$. Fix $\mathbb{P}$, let F_T be the cdf of $T(X)$ under $\mathbb{P}$. Since $X' \sim X$ under $\mathbb{P}$,

$$\mathbb{P}(T(X') \leq T(X) \mid X) = F_T(T(X)).$$

A standard probability fact: for any real random variable Z with cdf F , $\mathbb{P}(F(Z) \leq \alpha) \leq \alpha$. Applying this with $Z = T(X)$, $F = F_T$ gives $\mathbb{P}(P \leq \alpha) \leq \mathbb{P}(F_T(T(X)) \leq \alpha) \leq \alpha$, so P is a p-variable.

- ② $(i) \Rightarrow (ii)$. Take $T(X) := P$ (treat P itself as the statistic). Let F_P be P 's cdf under $\mathbb{P}$. Since P is a p-variable, by definition $F_P \leq \text{id}$ on $[0, 1]$, so

$$P \geq F_P(P) = \mathbb{P}(T(X') \leq T(X) \mid X).$$

Takeaway. A p-value *is*, structurally, the (bound on the) conditional probability that a fresh replicate's test statistic is no more extreme than the one observed – exactly the textbook interpretation of a p-value, now made rigorous for arbitrary composite nulls.

Side-by-side: e-value vs. p-value, structurally

Chapter 14 at a glance: e-values vs. p-values

	e-value	p-value
Monotone role of X	E increasing in X	P decreasing in X
Natural bound (Prop. 14.1)	$E \leq 1/S(X; \mathbb{P})$	$P \geq S(X; \mathbb{P})$
Duality (Thm. 14.2)	$\sup_{E \in \mathcal{E}^X} E$	$= 1 / \inf_{P \in \mathcal{U}^X} P$
Structural representation	$E \leq \mathbb{E}^{\mathbb{P}}[dQ/d\mathbb{P} \mid X]$ (Thm. 14.4)	$P \geq \mathbb{P}(T(X') \leq T(X) \mid X)$ (Thm. 14.5)
What it "is"	a conditional expectation of a likelihood ratio	a conditional probability of a rarer outcome
Existence of exact, powered version (Thm. 14.6)	$\text{Conv}(Q) \cap \text{Span}(\mathcal{P}) = \emptyset$	$\text{Conv}(\mathcal{Q}) \cap \text{Span}(\mathcal{P}) = \emptyset$ (same condition, via duality)

14.3 Setup: finite hypotheses, pivotal statistics

Intuition. Chapter 3 asked: when does a powered (nontrivial) e-value exist at all against a given alternative? Section 14.3 asks the sharper question: when does an *exact* e-value (or, equivalently, an exact p-value) that is also powered exist, when both null and alternative are finite collections of distributions?

Definitions

For a finite null $\mathcal{P} = \{\mathbb{P}_1, \dots, \mathbb{P}_L\}$, a random variable X is **pivotal** if it has the *same* distribution under every $\mathbb{P} \in \mathcal{P}$. Recall an e-variable is **exact** if $\mathbb{E}^{\mathbb{P}}[E] = 1$ for every $\mathbb{P} \in \mathcal{P}$ (then E is automatically pivotal-compatible), and a p-variable is **exact** if it is exactly $\text{Uniform}[0, 1]$ under every $\mathbb{P} \in \mathcal{P}$.

Being *powered against* a finite alternative $\mathcal{Q} = \{\mathbb{Q}_1, \dots, \mathbb{Q}_M\}$ means the e-value has expectation > 1 (resp. the p-value rejects with nontrivial probability) under every $\mathbb{Q} \in \mathcal{Q}$.

Theorem 14.6: the existence condition I

Theorem 14.6

Consider testing a finite $\mathcal{P} = \{\mathbb{P}_1, \dots, \mathbb{P}_L\}$ against a finite $\mathcal{Q} = \{\mathbb{Q}_1, \dots, \mathbb{Q}_M\}$. Under a regularity condition (U) permitting external randomization, the following are equivalent:

- (i) there exists an exact (hence pivotal) p-variable that is powered against $\mathcal{Q}$;
- (ii) there exists a pivotal, exact, bounded e-variable with positive e-power against $\mathcal{Q}$;
- (iii) there exists an exact e-variable that is powered against $\mathcal{Q}$;
- (iv) $\text{Conv}(\mathbb{Q}_1, \dots, \mathbb{Q}_M) \cap \text{Span}(\mathbb{P}_1, \dots, \mathbb{P}_L) = \emptyset$.

Theorem 14.6: the existence condition II

Span vs. convex hull

$$\text{Span}(\mathbb{P}_1, \dots, \mathbb{P}_L) = \left\{ \sum_{\ell=1}^L \lambda_{\ell} \mathbb{P}_{\ell} : \lambda_{\ell} \in \mathbb{R} \text{ for each } \ell \right\},$$

a set of *signed* measures, allowing *any real* coefficients λ_{ℓ} (not requiring $\lambda_{\ell} \geq 0$ or $\sum_{\ell} \lambda_{\ell} = 1$ as the convex hull would). *Symbols:* $\text{Conv}(\mathbb{Q}_1, \dots, \mathbb{Q}_M)$ is the usual convex hull (probabilistic mixtures) of the alternative; the null side is enlarged all the way to its linear span.

Theorem 14.6: the existence condition III

Why the span appears: step-by-step intuition for (iii) $\Rightarrow$ (iv).

- 1 Suppose an exact e-variable E is powered against $\mathcal{Q}$, so $\mathbb{E}^{\mathbb{Q}}[E] > 1$ for every $\mathbb{Q}$ in the convex hull $\text{Conv}(\mathcal{Q})$ (expectation is linear, so powered against the extreme points $\mathbb{Q}_1, \dots, \mathbb{Q}_M$ implies powered against all mixtures).
- 2 Suppose, for contradiction, some $\mathbb{Q}_0 \in \text{Conv}(\mathcal{Q})$ can also be written as a *signed* linear combination of the null measures,

$$\mathbb{Q}_0 = \sum_{\ell=1}^L \lambda_{\ell} \mathbb{P}_{\ell}, \quad \lambda_{\ell} \in \mathbb{R}.$$

Because $\mathbb{Q}_0$ is a probability measure and each $\mathbb{P}_{\ell}$ integrates to 1, applying both sides to the whole sample space forces $\sum_{\ell=1}^L \lambda_{\ell} = 1$.

- 3 Then, using linearity of expectation and *exactness* of E (i.e. $\mathbb{E}^{\mathbb{P}_{\ell}}[E] = 1$ for every ℓ),

$$\mathbb{E}^{\mathbb{Q}_0}[E] = \sum_{\ell=1}^L \lambda_{\ell} \mathbb{E}^{\mathbb{P}_{\ell}}[E] = \sum_{\ell=1}^L \lambda_{\ell} \cdot 1 = \sum_{\ell=1}^L \lambda_{\ell} = 1.$$

Theorem 14.6: the existence condition IV

- ④ This contradicts $\mathbb{E}^{\mathbb{Q}_0}[E] > 1$ from step 1. Hence no such signed representation $\mathbb{Q}_0 = \sum_{\ell} \lambda_{\ell} \mathbb{P}_{\ell}$ can exist, i.e. $\text{Conv}(\mathcal{Q}) \cap \text{Span}(\mathbb{P}_1, \dots, \mathbb{P}_L) = \emptyset$.

Theorem 14.6: the existence condition V

Comparison with Theorem 3.9. The basic existence result for (inexact) powered e-values in Chapter 3 required $\text{Conv}(\mathbb{Q}_1, \dots, \mathbb{Q}_M) \cap \text{Conv}(\mathbb{P}_1, \dots, \mathbb{P}_L) = \emptyset$ – disjointness from the *convex hull* of the null. Theorem 14.6 replaces $\text{Conv}(\mathcal{P})$ by the strictly larger $\text{Span}(\mathcal{P})$ (allowing negative and unnormalized coefficients), which is a *harder* condition to satisfy – reflecting the fact that requiring *exactness* (not just $\mathbb{E}^{\mathbb{P}}[E] \leq 1$) is a real restriction.

A full proof of Theorem 14.6 needs tools from optimal transport theory and is not reproduced here; only the intuitive direction (iii) $\Rightarrow$ (iv) is elementary, as shown above.

Chapter 14 takeaways

- **Duality (14.1).** For monotone transforms of one test statistic X , the best e-value and best p-value are exact reciprocals: $\sup_{E \in \mathcal{E}^X} E = 1 / \inf_{P \in \mathcal{U}^X} P$ $\mathbb{L}$ -a.s. (Theorem 14.2), sharpening Markov's inequality to a uniform, union-bound-free statement (Corollary 14.3).
- **Structural representations (14.2).** Every e-variable is dominated by a conditional expectation $\mathbb{E}^{\mathbb{P}}[dQ/d\mathbb{P} \mid X]$ of a likelihood ratio (Theorem 14.4); every p-variable is dominated by a conditional probability $\mathbb{P}(T(X') \leq T(X) \mid X)$ of a replicate being no more extreme (Theorem 14.5).
- **Existence with exactness (14.3).** For finite hypotheses, an exact, powered e-value (equivalently p-value) exists iff $\text{Conv}(\mathcal{Q}) \cap \text{Span}(\mathcal{P}) = \emptyset$ (Theorem 14.6) – the span, not the convex hull, of the null is the right geometric object once exactness is demanded.